

Project Design Phase-II

Technology Stack (Architecture & Stack)

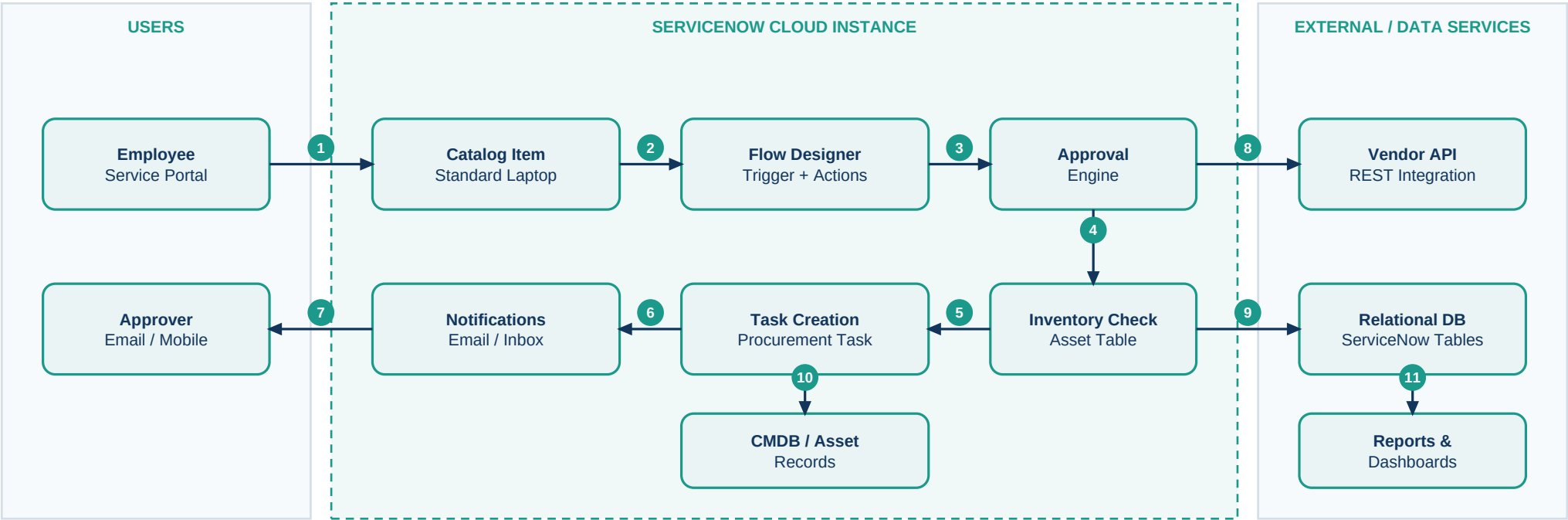
Date	30 September 2026
Team ID	6ab7efd38c7c66e706728f94
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Team

Role	Name
Team Lead	NAVINA K
Team Member	R Aakash
Team Member	A Mahalakshmi
Team Member	Maheswaran S
Team Member	Sabarinathan

Technical Architecture

Automated standard laptop ordering: an employee raises a request from the service portal, and a Flow Designer flow handles approval, inventory check, procurement task creation and notification without manual intervention.



Flow: 1 Request submitted → 2 Flow triggered → 3 Approval requested → 4 Inventory checked → 5 Procurement task created → 6 Notification sent → 7 Employee/approver updated → 8 Vendor order placed → 9 Records stored → 10 Asset records updated → 11 Reports refreshed.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees order a standard laptop through a service catalog item on the portal; approvers respond through email or mobile.	ServiceNow Service Portal, HTML, CSS, JavaScript
2	Application Logic-1	Flow Designer flow triggered when the catalog request is submitted; validates the request and routes it.	ServiceNow Flow Designer
3	Application Logic-2	Approval logic for manager and IT approval before ordering.	Flow Designer Approval action, Approval Engine
4	Application Logic-3	Procurement task creation, inventory check and notifications to requester and fulfilment team.	Flow Designer Actions, Notification Engine, GlideRecord scripts
5	Database	Requests, tasks, approvals and catalog data stored as relational tables.	ServiceNow platform database (relational)
6	Cloud Database	Platform-managed database hosted with the cloud instance.	ServiceNow Cloud Instance
7	File Storage	Attachments such as quotes, purchase orders and invoices.	ServiceNow Attachment Storage
8	External API-1	Sends the laptop order to the vendor and receives order status.	Vendor REST API via IntegrationHub
9	External API-2	Email delivery for approval and status notifications.	SMTP / Email Server
10	Machine Learning Model	Not required for standard laptop ordering; rule-based automation is used.	Not applicable
11	Infrastructure (Server / Cloud)	Application deployment on cloud-hosted ServiceNow instance with development and production separation.	ServiceNow Cloud (PDI / Production Instance)

Table-2 : Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source libraries used for front-end styling and scripting on the portal.	Bootstrap, AngularJS (Service Portal widgets)
2	Security Implementations	Role-based access control, ACLs for records, approver authorization, encrypted communication and audit logging.	ServiceNow ACLs, RBAC, HTTPS/TLS, OWASP guidelines
3	Scalable Architecture	Multi-tier platform architecture (presentation, application, data) with reusable flows and subflows that scale to higher request volume.	3-tier architecture, Flow Designer subflows
4	Availability	Cloud hosting with redundant data centres and load-balanced application nodes provides high availability.	ServiceNow load balancing, data-centre failover
5	Performance	Asynchronous flow execution, indexed tables and cached catalog data keep request handling fast.	Async flow execution, database indexing, platform caching